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MAST20030: Differential Equations
Tutorial Solutions: Week 4

Solution to Question 1.

(a) Now, we can work out the Taylor series of $f(t) = \log(1-t)$ by finding the values of its derivatives at $t = 0$:

n	0	1	2	3	4	$\cdots$	n
$f^{(n)}(t)$	$\log(1-t)$	$-\frac{1}{1-t}$	$-\frac{1}{(1-t)^2}$	$-\frac{2}{(1-t)^3}$	$-\frac{6}{(1-t)^4}$	$\cdots$	$-\frac{(n-1)!}{(1-t)^n}$
$f^{(n)}(0)$	0	-1	-1	-2	-6	$\cdots$	$-(n-1)!$

$$\Rightarrow \text{ the Taylor series is: } \qquad P(t) = -\sum_{n=1}^{\infty} \frac{(n-1)!}{n!} t^n = -\sum_{n=1}^{\infty} \frac{t^n}{n}.$$

The radius of convergence may be deduced from the ratio test

$$\lim_{n\to\infty} \left| \frac{-1}{n+1} \Big/ \frac{-1}{n} \right| = \lim_{n\to\infty} \frac{n}{n+1} = 1;$$

it is $1^{-1} = 1$.

(b) Because the derivative of $\log(1-t)$ is $-\dfrac{1}{1-t}$, it is the unique solution to the IVP

$$y' = -\frac{1}{1-t}, \qquad y(0) = 0.$$

Now, if $P(t)$ denotes the Taylor series computed in 1, then we have $P(0) = 0$ and

$$P'(t) = -\frac{\mathrm{d}}{\mathrm{d}t} \sum_{n=1}^{\infty} \frac{t^n}{n} = -\sum_{n=1}^{\infty} \frac{\mathrm{d}}{\mathrm{d}t} \frac{t^n}{n} = -\sum_{n=1}^{\infty} t^{n-1} = -\sum_{m=0}^{\infty} t^m.$$

This is a geometric series in t, hence the series converges to $\dfrac{1}{1-t}$ for all $|t| < 1$. In other words, $P(t)$ satisfies the same IVP as $\log(1-t)$, for $|t| < 1$, so we may conclude that

$$\sum_{j=1}^{\infty} \frac{t^n}{n} = \log(1-t), \qquad |t| < 1.$$

Solution to Question 2.

(a) Substituting gives the equation

$$\sum_{n=0}^{\infty} \left[n(n-1)c_n t^{n-2} - n c_n t^n - c_n t^n \right] = 0 \;\Rightarrow\; \sum_{n=2}^{\infty} n(n-1)c_n t^{n-2} = \sum_{n=0}^{\infty} (n+1)c_n t^n.$$

If we now shift the indices so that the powers of t are both t^n, we obtain

$$\sum_{n=0}^{\infty} (n+1)(n+2)c_{n+2} t^n = \sum_{n=0}^{\infty} (n+1)c_n t^n \;\Rightarrow\; c_{n+2} = \frac{c_n}{n+2}, \qquad n \geqslant 0.$$

This is the desired relation.

(b) We solve iteratively in order to see the pattern:

$$c_2 = \frac{1}{2}c_0, \qquad c_4 = \frac{1}{4}c_2 = \frac{1}{2\cdot 4}c_0, \qquad c_6 = \frac{1}{6}c_4 = \frac{1}{2\cdot 4\cdot 6}c_0, \qquad \cdots$$
$$c_3 = \frac{1}{3}c_1, \qquad c_5 = \frac{1}{5}c_3 = \frac{1}{3\cdot 5}c_1, \qquad c_7 = \frac{1}{7}c_5 = \frac{1}{3\cdot 5\cdot 7}c_1, \qquad \cdots$$
$$\Rightarrow\ c_{2m} = \frac{c_0}{2\cdot 4\cdot 6\cdots(2m)}, \qquad c_{2m+1} = \frac{c_1}{1\cdot 3\cdot 5\cdots(2m+1)}.$$

In principle, this should be done by induction, but we aren't insisting on that in this subject.

(c) We can obtain two linearly independent solutions by choosing, *eg.*, $(c_0, c_1) = (1, 0)$ and $(0, 1)$. The solutions obtained in this way are

$$y_1(t) = \sum_{m=0}^{\infty} \frac{t^{2m}}{2\cdot 4\cdot 6\cdots(2m)}, \qquad y_2(t) = \sum_{m=0}^{\infty} \frac{t^{2m+1}}{1\cdot 3\cdot 5\cdots(2m+1)}.$$

Their radii of convergence are both infinite because

$$\lim_{m\to\infty}\left|\frac{2\cdot 4\cdot 6\cdots(2m)}{2\cdot 4\cdot 6\cdots(2m)(2m+2)}\right| = \lim_{m\to\infty}\frac{1}{2m+2} = 0, \quad \lim_{m\to\infty}\left|\frac{1\cdot 3\cdot 5\cdots(2m+1)}{1\cdot 3\cdot 5\cdots(2m+1)(2m+3)}\right| = \lim_{m\to\infty}\frac{1}{2m+3} = 0.$$

Solution to Question 3.

Series in Question 1: we must demonstrate that that we can differentiate the series term by term. Using the theorem 'Differentiating a series' we must check that both the original series, and the term by term differentiated series, converge uniformly. Let $w_n(t) = \dfrac{t^n}{n}$, so that $\dfrac{\mathrm{d}w_n}{\mathrm{d}t} = t^{n-1}$. Let $0 < L < 1$, now $\max\limits_{t\in[-L,L]}\left|\dfrac{t^n}{n}\right| \leqslant L^{n-1}$ and $\max\limits_{t\in[-L,L]}|t^{n-1}| \leqslant L^{n-1}$ and since $\sum\limits_{n=1}^{\infty} L^{n-1}$ converges, both $\sum\limits_{n=1}^{\infty} w_n$ and $\sum\limits_{n=1}^{\infty}\dfrac{\mathrm{d}w_n}{\mathrm{d}t}$ converge uniformly on $[-L, L]$ by the Weierstrass test for uniform convergence. Therefore so we have satisfied the requirements for the differentiating a series theorem, i.e.,

$$\frac{\mathrm{d}}{\mathrm{d}t}\sum_{n=1}^{\infty} w_n(t) = \sum_{n=1}^{\infty}\frac{\mathrm{d}w_n}{\mathrm{d}t}.$$

Series in Question 2: we must demonstrate that we can twice-differentiate the series term by term. We will only demonstrate this for $y_1(t)$ but a similar analysis holds for $y_2(t)$. Let $w_m(t) = \dfrac{t^{2m}}{2\cdot 4\cdot 6\cdots(2m)}$ so that $\dfrac{\mathrm{d}w_m}{\mathrm{d}t} = \dfrac{t^{2m-1}}{2\cdot 4\cdot 6\cdots(2m-2)}$ and $\dfrac{\mathrm{d}^2w_m}{\mathrm{d}t^2} = \dfrac{(2m-1)t^{2m-2}}{2\cdot 4\cdot 6\cdots(2m-2)}$. Now let $a_m = \dfrac{(2m-1)L^{2m-2}}{2\cdot 4\cdot 6\cdots(2m-2)}$ and so $\max\limits_{t\in[-L,L]}\left|\dfrac{\mathrm{d}^2w_m}{\mathrm{d}t^2}\right| \leqslant a_m$, $\max\limits_{t\in[-L,L]}\left|\dfrac{\mathrm{d}w_m}{\mathrm{d}t}\right| \leqslant a_m$ and $\max\limits_{t\in[-L,L]}|w_m| \leqslant a_m$ where $0 < L$. We can use the ratio test to show that $\sum\limits_{m=1}^{\infty} a_m$ converges and so $\sum\limits_{m=1}^{\infty}\dfrac{\mathrm{d}^2w_m}{\mathrm{d}t^2}$, $\sum\limits_{m=1}^{\infty}\dfrac{\mathrm{d}w_m}{\mathrm{d}t}$ and $\sum\limits_{m=1}^{\infty} w_m$ converge uniformly on $[-L, L]$. Satisfaction of the differentiating a series theorem for both the first and second derivative ensures that

$$\frac{\mathrm{d}^2}{\mathrm{d}t^2}\sum_{n=1}^{\infty} w_n(t) = \sum_{n=1}^{\infty}\frac{\mathrm{d}^2w_n}{\mathrm{d}t^2}.$$

Solution to Question 4.

We only prove for a power series about $t = 0$ but it can be generalised for a power series about $t = a$. Consider the power series $\sum\limits_{n=0}^{\infty} c_n t^n$ with radius of convergence $r > 0$. It can be shown that the series then

also converges *absolutely* within the radius of convergence, i.e. $\sum_{n=0}^{\infty} |c_n t^n|$ converges for all $|t| < r$ (this was not discussed in the lecture). Choose l such that $0 < l < r$ and let $D = [-l, l]$. Now

$$\max_{t \in [-l,l]} |c_n t^n| \leqslant |c_n|\, l^n$$

and since $t = l$ lies within the radius of convergence, $\sum_{n=0}^{\infty} |c_n|\, l^n$ must converge and hence $\sum_{n=0}^{\infty} c_n t^n$ converges uniformly on $[-l, l]$.